

Eugenio Landi

Personal information

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Current position

September 2025 – **Postdoctoral researcher**
Now Dipartimento di Matematica – Guido Castelnuovo, Università degli studi di Roma “La Sapienza”

Past positions

July 2024 – **Postdoctoral researcher**
June 2025 Dipartimento di Matematica – Guido Castelnuovo, Università degli studi di Roma “La Sapienza”
June 2023 – **Postdoctoral researcher**
May 2024 Dipartimento di Matematica – Guido Castelnuovo, Università degli studi di Roma “La Sapienza”
August 2022 – **Adjunct researcher**
December 2022 Eberly College of Science, Department of Mathematics, Pennsylvania State University

Education

2018 – 2022 **Ph.D. in Mathematics** – Università di Roma Tre, *cum laude*
Thesis: “Equivariant localization methods, orientations and modularity”
Supervisor: Domenico Fiorenza
2016 – 2018 **Master Degree in Mathematics** – Università degli studi di Roma “La Sapienza”, 110/110 *cum laude*
Thesis: “The Landweber Exact Functor Theorem and Elliptic Cohomology”
Supervisor: Domenico Fiorenza
2013 – 2016 **Bachelor Degree in Mathematics** – Università degli studi di Roma “La Sapienza”, 110/110 *cum laude*
Thesis: “Il Gruppo di Cremona” (“The Cremona Group”)
Supervisor: Kieran O’Grady

Papers and preprints

- M. Coloma, D. Fiorenza, E. Landi, “*An exposition of the topological half of the Grothendieck–Hirzebruch–Riemann–Roch theorem in the fancy language of spectra*”, *Expositiones Mathematicae*.
<https://doi.org/10.1016/j.exmath.2021.11.003>, arXiv:1911.12035

- M. Coloma, D. Fiorenza, E. Landi, “*A very short note on the (rational) graded Hori map*”, Communications in Algebra.
<https://doi.org/10.1080/00927872.2021.2005076>,
arXiv:2003.13066
- M. Coloma, D. Fiorenza, E. Landi, “*The (anti-)holomorphic sector in $\mathbb{C}/\Lambda$ -equivariant cohomology, and the Witten class*”, Journal of Geometry and Physics. <https://doi.org/10.1016/j.geomphys.2023.104750>,
arXiv:2106.14945
- D. Fiorenza, E. Landi, “*Integrals detecting degree 3 string cobordism classes*”, Bollettino dell’Unione Matematica Italiana.
<https://doi.org/10.1007/s40574-023-00378-4>, arXiv:2209.12933
- G. Cerulli Irelli, D. Fiorenza, E. Landi, M. Matteucci, “*A categorification of Kauffman states for planar graphs*”, arXiv:2605.19872
- E. Landi, “*An Abstract Index Theorem via Rees Algebras*”,
arXiv:2608.30587

Teaching experiences

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| 2019 – 2020 | Teaching Assistant – Geometry, Dept. of Basic and Applied Sciences for Engineering (SBAI), Università degli studi di Roma “La Sapienza”, Clinical engineering |
| 2019 – 2020 | Teaching Assistant – Geometry, Dept. of Basic and Applied Sciences for Engineering (SBAI), Università degli studi di Roma “La Sapienza”, Building engineering |
| August 2022 –
December 2022 | Lecturer – Math220: Matrices, Eberly College of Science, Pennsylvania State University |
| March 2026 | Ph.D. Course – Model Categories and Abstract Homotopy Theory, Dipartimento di Matematica – Guido Castelnuovo, Università degli studi di Roma “La Sapienza” |

Invited talks

- The Witten genus as the “volume” of the double loop space, 18/03/2021, KCL/UCL Junior Geometry seminar, London.
- Equivariant localization methods, orientations and modularity, 19/04/2022, GAP (Geometry, Algebra, Physics) Seminar, Pennsylvania State University.
- Lecture on Model Categories, 10/05/2022, Preparatory informal seminar on the Geometric Cobordism Hypothesis.
- Equivariant localization methods, orientations and modularity, 26/05/2022, First UMI meeting of PhD students, Padova.
- String bordism invariants in dimension 3 from $U(1)$ -valued TQFTs, 13/10/2022, Cobordisms, Strings, and Thom Spectra, Oaxaca.
- Breve Seminario sugli Spettri (A Short Talk on Spectra), 04/11/2022, Lecture Series on Cyclic Homology, Rome – Tor Vergata.
- The Topological Half of the Grothendieck–Hirzebruch–Riemann–Roch Theorem, 09/11/2022, Cohomology in algebra, geometry, physics and statistics, Prague.

- The Topological Half of the Grothendieck–Hirzebruch–Riemann–Roch Theorem, 01/12/2022, Workshop on “Higher Structures” at Penn State, Pennsylvania State University.
- String bordism invariants in dimension 3 from $U(1)$ -valued TQFTs, 06/12/2022, GAP (Geometry, Algebra, Physics) Seminar, Pennsylvania State University.
- String bordism invariants in dimension 3 from $U(1)$ -valued TQFTs, 17/08/2023, Higher Structures in Functorial Field Theory, Regensburg.
- Wrong-way maps in cyclic homology, 10/12/2024, GAP (Geometry, Algebra, Physics) Seminar, Pennsylvania State University.

Refereeing and reviews

- Referee for *Journal of Geometry and Physics*.
- Reviewer for ZBMath.

Grants

- Winner of a grant “Progetti di Avvio alla Ricerca – tipo 2”, 2024, Università degli studi di Roma – La Sapienza, for the project “Mappe wrong-way in omologia ciclica”.
- Winner of a grant “Progetti di Avvio alla Ricerca – tipo 2”, 2025, Università degli studi di Roma – La Sapienza, for the project “Deformations and higher structures”.

Visits

1/12/2024 – **Pennsylvania State University**
 13/12/2024 Short visit to Ping Xu and Mathieu Stiénon.